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PALEOMAGNETIC AND $^{40}Ar/^{39}Ar$ EVIDENCE FOR REMAGNETIZATION OF MESOZOIC OCEANIC ROCKS ON THE VIZCAINO PENINSULA, BAJA CALIFORNIA SUR, MEXICO

Jonathan T. Hagstrum
U.S. Geological Survey, Menlo Park, California, U.S.A.

Margarita López Martínez
Centro de Investigación Científica y de Educación Superior de Ensenada, B.C., México

Derek York
Department of Physics, University of Toronto, Toronto Ontario, Canada

Abstract. Previously published paleomagnetic data for Upper Jurassic pillow lavas of the Vizcaino Peninsula indicate that they were deposited near a paleolatitude of 14°N or S. Whether or not this result implies northward transport with respect to the continental interior has been controversial due to the lack of reliable Jurassic reference poles for the North American plate. Available paleomagnetic data for nearby Upper Triassic pillow basalts and overlying pelagic sediments at Punta San Hipólito, however, fail a fold test indicating that these rocks were remagnetized post-folding. Indistinguishable in-situ paleomagnetic directions and perturbed $^{40}Ar/^{39}Ar$ age spectra for the Triassic and Jurassic pillow lavas are consistent with resetting of their magnetic and isotopic systems in the middle Cretaceous, probably during burial by the overlap Valle Formation (>10 km thick). Resetting apparently occurred post-deformation so the paleolatitude of remagnetization is unknown. High-coercivity directions from a few samples of the Triassic rocks might represent an older magnetization acquired during deposition at paleolatitudes near 6°N or S.

Introduction

Accreted oceanic rocks of Mesozoic age are distributed along the western margin of North America from México to Alaska. These rocks have a complex post-accretion geologic history, often including burial and uplift, alteration and metamorphism, structural deformation and dismemberment, and in many cases significant coastwise transport [*Jarrard*, 1986] with respect to stable North America. Paleogeographic reconstructions of accreted oceanic terranes along the western coast of North America depend primarily on paleomagnetic data, which generally indicate southerly paleolatitudes of origin and subsequent northward transport consistent with plate motion models [*Beck*, 1989]. *Butler et al.* [1991] alternatively suggest that paleomagnetic results for Jurassic pillow basalts at Stanley Mountain in southern California [*McWilliams and Howell*, 1982] and on the Vizcaino Peninsula in Baja California Sur [*Hagstrum et al.*, 1985] show they did not undergo significant northward displacement as previously proposed, because revised reference paleomagnetic poles imply that North America was at lower paleolatitudes during the Middle Jurassic as well. These early interpretations have assumed, however, that the paleomagnetic data for the Stanley Mountain and Vizcaino Peninsula (Eugenia Formation) basalts represent primary thermoremanent magnetizations.

Recently, a paleomagnetic study of Jurassic sedimentary rocks overlying the Coast Range ophiolite at Stanley Mountain has indicated that these rocks were remagnetized somtime after accretion and burial by the Great Valley sequence [*Hagstrum*, 1992]. Moreover, sequences of Mesozoic pillow basalt and red radiolarian chert on Cedros Island in Baja California [*Hagstrum and Sedlock*, 1992] have been shown to be extensively remagnetized, although primary components of remanent magnetization have also been preserved to varying degrees within the chert samples. These primary components indicate that the cherts were deposited within the equatorial zone of high biologic productivity and accreted to the American margin at low (<20°N) paleolatitudes. Although remagnetization appears to have generally occurred during or relatively soon after accretion of these rocks, based on stratigraphic and structural arguments, little direct evidence is available for the timing of remagnetization.

In this study, $^{40}Ar/^{39}Ar$ analyses were made on paleomagnetic core samples, previously analysed by *Hagstrum et al.* [1985], from coastal exposures of Upper Triassic and Upper Jurassic pillow basalts on the Vizcaino Peninsula in Baja California Sur to provide isotopic evidence that might bear on the timing of magnetization acquisition in these rocks. Based on an intrastratal fold test, the paleomagnetic results show that the Triassic La Costa ophiolite at San Hipólito was strongly, although not completely, overprinted. A second high-coercivity and high-temperature component in basalt and overlying chert samples was tentatively interpreted as indicating deposition near a paleolatitude of 19°N or S, and is reinterpreted here. Uniform-polarity characteristic magnetizations for the homoclinal section of Jurassic Eugenia pillow basalts were interpreted to indicate deposition near 14°N or S [*Hagstrum et al.*, 1985].

Geologic Setting

On the Vizcaino Peninsula (Figure 1), Mesozoic basement rocks have been subdivided into three fault-bounded terranes: (1) the Puerto Nuevo terrane, characterized by blueschist and other metamorphic rocks in mélange; (2) the Vizcaino Sur terrane, characterized by Triassic ophiolite and pelagic sedimentary rocks, Lower Jurassic volcaniclastic rocks, and Middle Jurassic to Lower Cretaceous intrusive rocks; and (3) the Vizcaino Norte terrane, characterized by Triassic ophiolite

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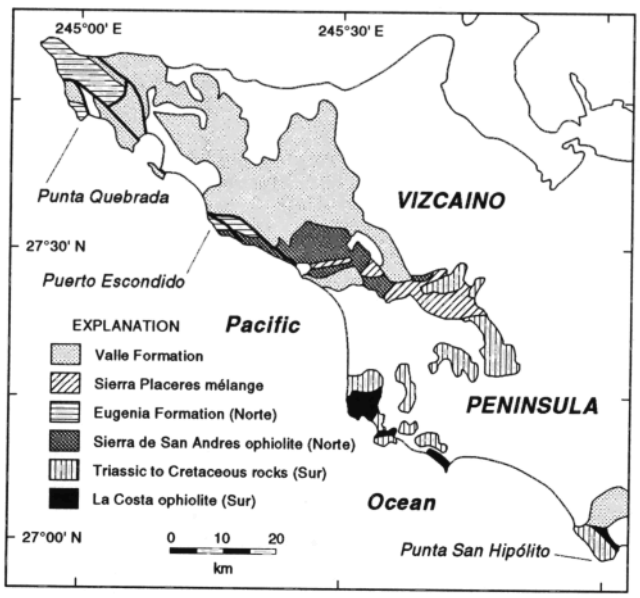


Fig. 1. Geologic map of the Vizcaino Peninsula in northwestern Baja California Sur, México, showing the distribution of Mesozoic rocks [after *Moore*, 1985]; unshaded areas denote Cenozoic rocks. (Norte), Vizcaino Norte terrane; (Sur), Vizcaino Sur terrane. Heavy lines mark fault contacts; lighter lines are depositional contacts.

and tuffaceous sedimentary rocks, and Upper Jurassic to Lower Cretaceous graywacke. The Vizcaino Norte and Sur terranes are separated by the Sierra Placeres serpentinite-matrix mélange which likely represents a major suture zone [*Moore*, 1985]. These two terranes and intervening suture are overlapped by middle Cretaceous sedimentary rocks, mainly turbidites, of the Valle Formation [*Patterson*, 1984].

The oldest rocks of the Vizcaino Sur terrane is the La Costa ophiolite exposed at Punta San Hipólito, and radiolaria from interpillow chert near its top are Late Triassic (Carnian-Norian) in age [*Barnes*, 1982]. The altered low-K basalts of the ophiolite are directly overlain by the San Hipólito Formation which consists in ascending order of chert, limestone, breccia, and sandstone members. The lower two members contain the megafossils *Monotis* and *Halobia* and a radiolarian assemblage that are also Carnian-Norian in age [*Pessagno et al.*, 1979]; the uppermost sandstone member contains radiolarians indicating an Early Jurassic age [*Whalen and Pessagno*, 1984]. These strata have been subsequently partially recrystallized and contain low-grade metamorphic minerals including albite, laumontite, quartz, epidote, and abundant carbonate minerals [*Moore*, 1985].

The Eugenia Formation is stratigraphically the uppermost unit of the Vizcaino Norte terrane, and consists largely of marine fan sediments, including volcanogenic conglomerate, graywacke, and shale, and minor intercalated volcanic rocks [e.g., *Barnes*, 1982]. A composite section of lower and upper parts [*Boles and Hickey*, 1979] totals ~2,500 m in thickness, and the lower part (1,500 m thick) contains megafossils and radiolarian assemblages that range in age from Late Jurassic (Tithonian) to Early Cretaceous (Valanginian). Secondary mineral assemblages within the sandstones have been interpreted by *Barnes* [1982] to indicate mild thermal diagenesis at low temperatures (~100°C). At Punta Quebrada and Puerto Escondido, the Eugenia Formation contains a sequence of clinopyroxene-plagioclase phyric pillow lavas as much as 650 m thick. These volcanic rocks locally retain glassy textures and show major and trace element chemistries indicating an island arc-tholeiitic composition [*Moore*, 1985]. The unconformable upper part of the Eugenia Formation consists predominantly of sandstones that contain Aptian-Albian foraminifera [*Boles and Hickey*, 1979].

Methods and Results

Paleomagnetic sampling and laboratory procedures for the Punta San Hipólito, Puerto Escondido, and Punta Quebrada collections are reported in *Hagstrum et al.* [1985]. The paleomagnetic data are also presented here in Table 1 and Figure 2, but with several changes. Least-squares lines have been refitted to the demagnetization data, so the characteristic site-mean directions and their statistical parameters are slightly different from those given in *Hagstrum et al.* [1985]. Also, samples from the La Costa ophiolite (LCB) and members of the San Hipólito Formation (SHC, SHL, SHS) have been grouped into individual sites (Table 1).

Demagnetization planes were incorrectly used by *Hagstrum et al.* [1985] to define a high-coercivity component for the La Costa basalt samples because the presence of a

TABLE 1. Paleomagnetic Data for Mesozoic Oceanic Rocks, Vizcaino Peninsula

Sites	I	D	N	R	k	α_{95}	λ	ϕ
Punta Quebrada (27.77°N, 245.04°E)								
PQ1	50.7	339.6	6	5.4858	10	22.6	71.9	171.6
PQ2	49.7	338.3	6	5.7045	17	16.8	70.9	168.5
PQ3	35.2	331.1	4	3.9554	67	11.3	62.3	143.8
Puerto Escondido (27.54°N, 245.27°E)								
PE1	49.1	326.0	5	4.8851	35	13.2	60.2	168.2
PE2	72.9	10.9	4	3.9544	66	11.4	58.2	256.1
PE3	76.8	59.3	5	4.9256	54	10.5	37.7	272.7
PE4	53.4	350.2	6	5.8829	43	10.4	79.4	195.0
PE5	62.9	334.8	6	5.8833	43	10.4	63.8	201.7
Avg	58.0	341.9	(8)	7.6562	20	12.6	71.3	195.8
Avg/c	31.3	358.9	(8)	7.6069	18	13.5	79.3	71.0
Punta San Hipólito (27.03°N, 246.00°E)								
SHS	65.1	345.2	10	9.7198	32	8.7	66.8	219.8
SHL	54.8	350.1	14	13.7761	58	5.3	78.2	202.9
SHC	42.7	348.2	10	9.7438	35	8.3	79.2	146.6
LCB	38.9	346.4	8	7.8044	36	9.4	76.6	136.7
Avg	50.3	347.6	(4)	3.9338	45	13.8	78.4	179.4
Avg/c	52.5	284.1	(4)	3.8920	28	17.8	26.3	181.6
PSH	-47.5	2.2	4	3.6798	9	31.7	34.3	63.7
PSH/c	-12.8	14.0	4	3.6798	9	31.7	53.8	42.0

I and D are inclination and declination of paleomagnetic directions (°) uncorrected (/c, corrected) for structural tilt; N is number of samples (sites) analyzed; R is vector sum of N unit vectors; k is the precision parameter; α_{95} is the radius of 95% confidence (°); λ and ϕ are latitude and east longitude of virtual geomagnetic pole (°).

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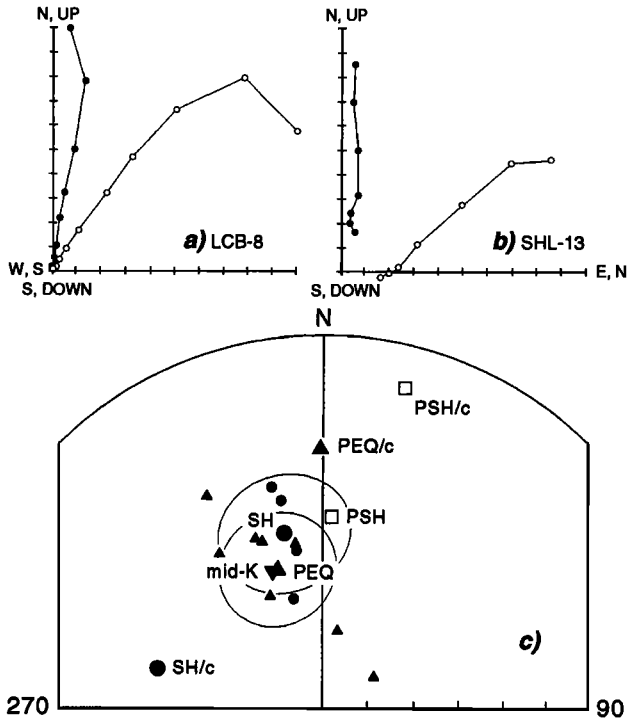


Fig. 2. Orthogonal projections of alternating field and thermal demagnetization curves for samples of (a) La Costa pillow basalt to 85 mT and (b) San Hipólito limestone to 500°C; solid (open) symbols indicate projection onto the horizontal (vertical) plane. Equal-area plot (c) of mean paleomagnetic directions for samples from the Eugenia Formation (triangles), and from Punta San Hipólito (circles; Table 1). Inverted triangle, mid-Cretaceous reference direction. Small symbols indicate site-mean directions and large symbols unit-mean (with 95% confidence limits) directions uncorrected (/c, corrected) for structural tilt; squares indicate primary(?) negative-inclination directions for Punta San Hipólito samples.

strong overprint component in these rocks tends to force the remagnetization circles to converge at a negative-inclination direction (HB12') near the antipode of the positive-inclination overprint direction. The HC1' and HC2' directions for the chert samples reported by [*Hagstrum et al.*, 1985] were selected because they appeared antipodal to the HB12' direction, and they probably represent a spurious intermediate direction between the overprint direction and an older unresolved direction. In this reevaluation, least-squares lines are fitted to negative-inclination directions for two basalt and two limestone samples from Punta San Hipólito (Figure 2a, 2b) and their average (PSH) is given in Table 1 (Figure 2c). Although demagnetization curves for the chert samples often missed the origin suggesting the presence of a second higher-temperature component of magnetization in these rocks as well, this component could not be fitted by least-squares lines.

Whole-rock basalt samples (~2 mm in diameter) from Punta Quebrada, Puerto Escondido, and Punta San Hipólito, were irradiated for 30 hr at a power of 2 mW in position 5c of the uranium enriched research reactor at McMaster University; samples of monitor hb 3gr were also irradiated in the same can. $^{40}Ar/^{39}Ar$ step-heating experiments were conducted with a fully automated argon laser-probe at the University of Toronto, and the purified gases were analysed with a mass spectrometer. Typically 7 or 8 fractions were collected between 600° to 1600°C, and the age spectra and Ca/K diagrams obtained are given in Figure 3.

Discussion and Conclusions

The $^{40}Ar/^{39}Ar$ age spectra for basalt samples collected from the La Costa ophiolite at Punta San Hipólito, and the Eugenia Formation at Puerto Escondido and Punta Quebrada, shown in Figure 3, are clearly perturbed. Based on fossil evidence, the expected age for the La Costa ophiolite is Late Triassic or ~220 Ma [*Harland et al.*, 1989]. The first gas fraction released for this sample (Figure 3a) yields virtually a zero age, but the remainder of the gas (~88% of ^{39}Ar released) defines a pseudo-plateau at ~110 Ma. Overprinting of the La Costa ophiolite and San Hipólito Formation was demonstrated by an intrastratal fold test within the limestone member [*Hagstrum et al.*, 1985], and remagnetization is probably related to alteration and partial recrystallization of these rocks at low temperatures which also might have caused resetting of the K-Ar isotopic system.

Similarly, the expected age for the pillow lavas of the Eugenia Formation (latest Jurassic or ~150 Ma to Valanginian

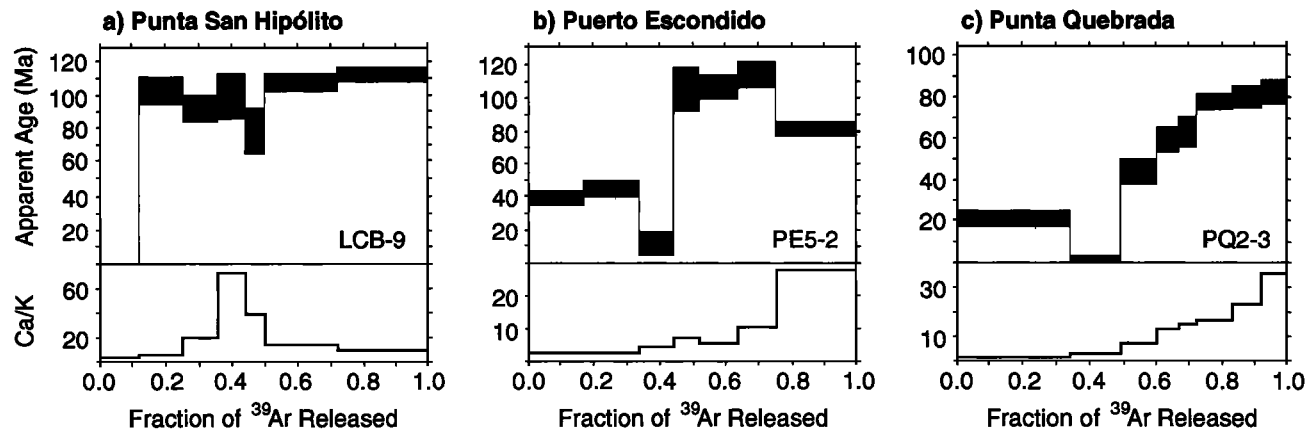


Fig. 3. $^{40}Ar/^{39}$ Ar whole-rock age spectra and Ca/K ratios for samples from Upper Triassic pillow basalt at (a) Punta San Hipólito, and Upper Jurassic pillow basalt at (b) Puerto Escondido and (c) Punta Quebrada. Two experiments were performed on Punta Quebrada samples yielding reproducible age spectra, and for simplicity the results from only one experiment is shown.

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~130 Ma) is much older than the apparent ages observed in these rocks. The age spectra for the Puerto Escondido and Punta Quebrada samples (Figure 3b and 3c) indicate recent argon loss as well, and are more disturbed than the La Costa spectra, possibly because of continued alteration. Although the Eugenia age spectra imply minimum ages of ~80-110 Ma, they do not necessarily indicate remagnetization of these rocks during middle Cretaceous time. Mean in-situ paleomagnetic directions for the Punta San Hipólito and Eugenia rocks (Figure 2c), however, are statistically indistinguishable from one another (and from a middle Cretaceous reference direction [*Globerman and Irving*, 1988]) suggesting that acquisition of the overprint magnetization in the San Hipólito rocks and of the characteristic magnetization in the Eugenia pillow lavas was coeval and post-deformational.

Deposition of the overlap Valle Formation beginning in Albian time [*Patterson*, 1984] indicates that the Vizcaino Norte and Sur terranes were accreted to the American margin by the middle Cretaceous. One possible interpretation of the paleomagnetic and $^{40}Ar/^{39}Ar$ data is that the magnetic and isotopic systems were reset during burial of these rocks by the Valle Formation (>10 km thick) after accretion. We infer, therefore, that the pillow basalts of the Vizcaino Norte and Sur terranes were remagnetized in middle Cretaceous time after they had been juxtaposed and deformed, probably within a subduction zone. Because structural deformation preceeded remagnetization, paleohorizontal and hence the paleolatitude at which these rocks were accreted to the continental margin cannot be determined. The negative inclination directions observed in a few samples from Punta San Hipólito (Table 1), however, could be tentatively interpreted as vestiges of a primary magnetization acquired during deposition of these rocks at low paleolatitudes near 6°N or S.

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J. T. Hagstrum, U.S. Geological Survey, 345 Middlefield Road, MS 937, Menlo Park, CA 94025, U.S.A.

M. López Martínez, Centro de Investigación Científica y de Educación Superior de Ensenada, Apdo. postal 2732, Ensenada, B.C., México. Mailing address: CICESE-Geología, P.O. Box 434843, San Diego, CA 92143-4843, U.S.A.

D. York, Department of Physics, University of Toronto, Toronto Ontario, M5S 1A7 Canada.

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